

REDOX

1. Strongest reducing agent among the following is
(1) K (2) Mg (3) Al (4) Ba
2. Which is not a Redox reaction
(1) $\text{CaCO}_3 \longrightarrow \text{CaO} + \text{CO}_2$
(2) $2\text{H}_2 + \text{O}_2 \longrightarrow 2\text{H}_2\text{O}$
(3) $\text{Na} + \text{H}_2\text{O} \longrightarrow \text{NaOH} + 1/2 \text{H}_2$
(4) $\text{MnCl}_3 \longrightarrow \text{MnCl}_2 + 1/2 \text{Cl}_2$
3. In a conjugate pair of reductant and oxidant, the oxidant has
(1) Higher oxidation number
(2) Lower oxidation number
(3) Same oxidation number
(4) None of these
4. A sulphur containing species that can not be act as oxidising agent is
(1) H_2SO_4 (2) H_2S
(3) SO_2 (4) H_2SO_3
5. In which is transfer of five electrons takes place
(1) $\text{MnO}_4^- \longrightarrow \text{Mn}^{+2}$ (2) $\text{CrO}_4^{2-} \longrightarrow \text{Cr}^{+3}$
(3) $\text{MnO}_4^- \longrightarrow \text{MnO}_2$ (4) $\text{Cr}_2\text{O}_7^{2-} \longrightarrow 2\text{Cr}^{+3}$
6. In which of the following compound N has lowest oxidation number
(1) NO (2) NO_2 (3) N_2O (4) N_2O_5
7. Nitrogen has fractional oxidation number in
(1) N_2H_4 (2) NH_4^+ (3) HN_3 (4) N_2F_2
8. Which of the following can have both positive and negative oxidation states
(1) F (2) I (3) Na (4) He
9. In balancing the half reaction $\text{S}_2\text{O}_3^{2-} \longrightarrow \text{S(s)}$ the number of electrons that must be added is
(1) 2 on the right
(2) 2 on the left
(3) 3 on the right
(4) 4 on the left
10. 0.1 M solution of KMnO_4 (in acidic medium) may oxidise
(1) 0.25 M $\text{C}_2\text{O}_4^{2-}$ (2) 0.8 M Fe^{+3}
(3) 0.1 M SO_4^{2-} (4) 0.6 M $\text{Cr}_2\text{O}_7^{2-}$
11. In KMnO_4 , K_2MnO_4 , MnO_2 and Mn_2O_3 compound in which element Mn shows its lowest oxidation number is
(1) KMnO_4 (2) K_2MnO_4
(3) MnO_2 (4) Mn_2O_3
12. Standard reduction electrode potentials of 3 metals A, B, C are +5V, -3V and -1.2V respectively. The reducing power of these metals are—
(1) $\text{B} > \text{C} > \text{A}$ (2) $\text{A} > \text{B} > \text{C}$
(3) $\text{C} > \text{B} > \text{A}$ (4) $\text{A} > \text{C} > \text{B}$
13. A reagent which can increase the oxidation number of an element in a given substance is called as—
(1) oxidant (2) reductant
(3) proton acceptor (4) proton donor